REVEAL: Agentic In-Hand Manipulation for Language-Guided Target Presentation

Anonymous CVPR submission

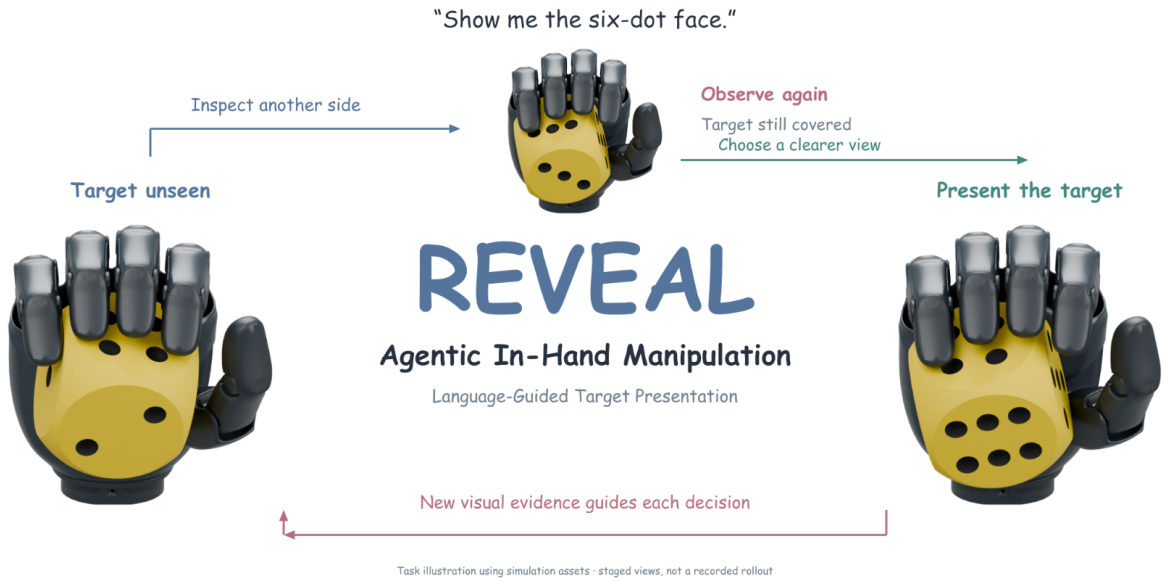


Figure 1. **REVEAL overview.** New visual evidence guides semantic decisions as the hand inspects and presents a requested object face. This staged simulation illustration is not a recorded rollout or physical-robot result.

Abstract

Language-guided in-hand manipulation requires a robot to uncover a requested object face while maintaining contact. This couples visual information gathering with motor control, yet high-level visual reasoning introduces delays during which the hand must continue acting. We present REVEAL, an asynchronous framework for online visual search and semantic target presentation. The agent uses updated images to inspect the object and bind the requested target to a shared goal bank, which interfaces with a history-conditioned motor policy. Teacher-supervised local controllers adapt joint commands during pending requests and subsequent execution. Simulation evaluations cover six object-specific systems, with additional English/Chinese instruction tests on three objects. Under changed semantic mappings, REVEAL achieves an object-averaged success rate of 65.8%, exceeding the strongest evaluated baseline by 40.4 percentage points.

1. Introduction

A command such as "present the face marked T" specifies a visual goal, not a sequence of hand motions. The robot must find the mark, associate it with a controllable orientation, and rotate the object while preserving contact. The target may be hidden by the object or the fingers, so an informative action also changes the grasp and the next observation. In-hand target presentation therefore couples visual reasoning with contact-rich control.

Vision-language agents offer a flexible way to interpret such requests, but their decisions arrive asynchronously. During inference, the hand and object continue to move. Repeating the previous command may lose the grasp or leave the target occluded; slowing motion alone may preserve contact without producing a useful correction. A practical system needs a local controller that responds to current measurements while semantic reasoning is pending.

Our key insight is to separate the rate of semantic decisions from the rate of motor control while keeping both interfaces grounded in the same goal representation. An

asynchronous visual agent inspects the object and issues a validated semantic command. A history-conditioned student maps the active goal and measured pose to joint targets. For the cube, separate local controllers adapt those targets during pending requests and ordinary execution, using privileged teacher actions for training. This lets the hand respond to its current state without waiting for another high-level call.

We instantiate this design by adapting a ViserDex-style RGB-to-pose policy [4] to a continuous semantic goal input while reusing compatible recurrent weights. The REVEAL agent emits public tool arguments that map to the shared goal bank; it does not expose hidden model embeddings. A fine-tuned Qwen3-VL baseline [2] predicts a goal vector directly, and a rule-based VLM parses a color and readability decision. These interfaces provide a common motor policy while retaining the differences between an agent with feedback and history and single-step grounding baselines.

Across 1,440 simulation trials spanning six object-specific systems and two semantic mappings, REVEAL attains object-averaged success rates of 70.4% on original mappings and 65.8% on changed mappings. These exceed the strongest evaluated baseline mean in each setting by 26.2 and 40.4 percentage points, respectively. The means describe complete systems with object-specific weights and perception, not zero-shot policy transfer. A separate 2,160-trial English/Chinese study favors REVEAL over fine-tuned grounding in all eight duck and hedgehog conditions after multiplicity correction; cube differences remain unresolved. Matched cube interventions separately test fresh visual evidence, inspection policy and local control.

Our contributions are threefold. First, we formulate language-guided target presentation as online visual search and semantic binding during continuous in-hand control. Second, we develop an asynchronous agent–motor interface with public semantic actions and teacher-supervised local adaptation. Third, we evaluate that interface through six-object mapping comparisons, three-object language coverage, and cube interventions that examine fresh visual evidence and control during pending requests.

Task mappings, model weights, stopping rules, and physical success criteria are fixed within each cohort. Service exclusions are applied symmetrically within comparisons, with all raw outcomes retained. We report physical reach separately from correct semantic binding and autonomous completion.

2. Related Work

RL for dexterous in-hand manipulation. Simulation-trained RL policies enable physical-hand reorientation with domain randomization [1] and general reorientation across diverse object geometries [7]. Rapid motor adaptation supports fingertip rotation across changing object properties [15]; RotateIt fuses vision, touch and proprioception through teacher–student distillation [16]. AnyRotate extends tactile control to arbitrary rotation axes and hand orientations [21]. These works establish motor capabilities on which semantic interaction can build. REVEAL studies how to discover and bind a language-specified target when its appearance is initially unavailable, rather than a new general-purpose reorientation algorithm.

Visual and goal-conditioned dexterity. ViserDex combines monocular pose estimation with recurrent control [4]; we reuse this decomposition and adapt a released Revo3 policy to semantic goals. GoalDex conditions a visuomotor policy on current and goal images, uses a VLM to select goals from language, and studies multi-object sim-to-real transfer [3]. It also models visual-processing delays. REVEAL focuses on acquiring missing evidence and binding the requested face during manipulation while local control continues. ManipTrans studies residual learning for dexterous bimanual transfer [12]; our local residual adaptation supports this semantic interaction loop.

Agentic reasoning and embodied feedback. SayCan grounds language-level skill selection in robotic affordances [11], while Inner Monologue incorporates environmental feedback into closed-loop planning [9]. Code as Policies composes perception and control APIs [14]; VoxPoser and ReKep ground language in spatial value maps and keypoint constraints [8, 10]. These works motivate separating semantic decisions from motor execution. Magma connects multimodal representations to agent actions [20], and VERA includes language-conditioned dexterous reorientation from visual plans [13]. REVEAL concentrates on an occluded target held in the hand: inspection changes available evidence, binding selects a canonical goal, and local controllers maintain manipulation during external reasoning. Our fresh-image intervention tests the role of later observations; feedback-driven planning itself is not a novelty claim.

Asynchronous robot policies. Real-time chunk execution [5], online VLA corrections [18], latency-aware RL [22], and temporal counterfactual preferences [23] address inference delays. LiMA combines sparse semantic intention with asynchronous dexterous action generation [6]. REVEAL learns compact teacher-supervised controllers around a separate motor policy and public semantic tools. Its evaluated distinction is the observation–binding–control loop; asynchronous computation and residual actions alone are not novel. The controller ablations do not isolate the benefit of explicit goal conditioning.

3. Task and Evaluation Contract

Semantic target. Q1 covers six object-specific simulation releases: a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton. Q2/Q3 interventions use only the marked cube. Cube and marked-object instructions identify one of six surface letters; die instructions request one through six dots. The cube letters are $\{T, E, R, L, S, H\}$. Each target has a canonical world orientation $Q(g)$ that places its marked face upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. All methods use the same scoring convention.

At step t, the agent observes RGB evidence I_t and the instruction ℓ. The motor system observes proprioception x_t and measured pose $\hat{p}_t$. The cube obtains $\hat{p}_t$ from RGB; the other releases use a noisy simulator-pose model. Clean true pose p_t and teacher actions are used for supervision and independent scoring, and future semantic responses are unavailable to online control. We denote the active goal vector by g_t, recurrent state by h_t, and the pending-request flag by m_t.

Physical time. One request may be pending at a time. The cube protocol allows six model calls per episode and 3,600 logical control steps at 30 Hz; the other objects use 20 Hz over the same 120-second logical horizon. If a call starts at k and takes L wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

Autonomous completion. All cube systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. The other object releases use calibrated completion correction for REVEAL and a measured-pose gate for bare baselines, with the same independently scored tolerance; their stopping conditions are documented in the supplement. At stop time τ, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

Matched reporting. Unless stated otherwise, each arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and scored physical targets. Instructions also match within the control and feedback studies. In the remapping study, the requested letter changes with the mapping to preserve the physical target. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

4. Method

REVEAL connects semantic decisions to motor control through a shared goal bank (Fig. 2). We detail the cube implementation below; other object releases use their own history-conditioned students, noisy pose feedback and maintenance actors, as specified in the supplement.

4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a 120×120 RGB crop. Calibration and rigid fitting recover the measured position and XYZW orientation, together with a fit residual. Cube trials use rendered object views with live hand occlusion; deployment receives no ground-truth pose.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

We fit the new goal interface and adapt the recurrent core with frozen-teacher actions in a DAgger-style procedure [17]. Teacher state and actions are training-only. The cube deployment uses the frozen RGB pose estimator; object-specific policy and perception differences for the other releases are documented in the supplement.

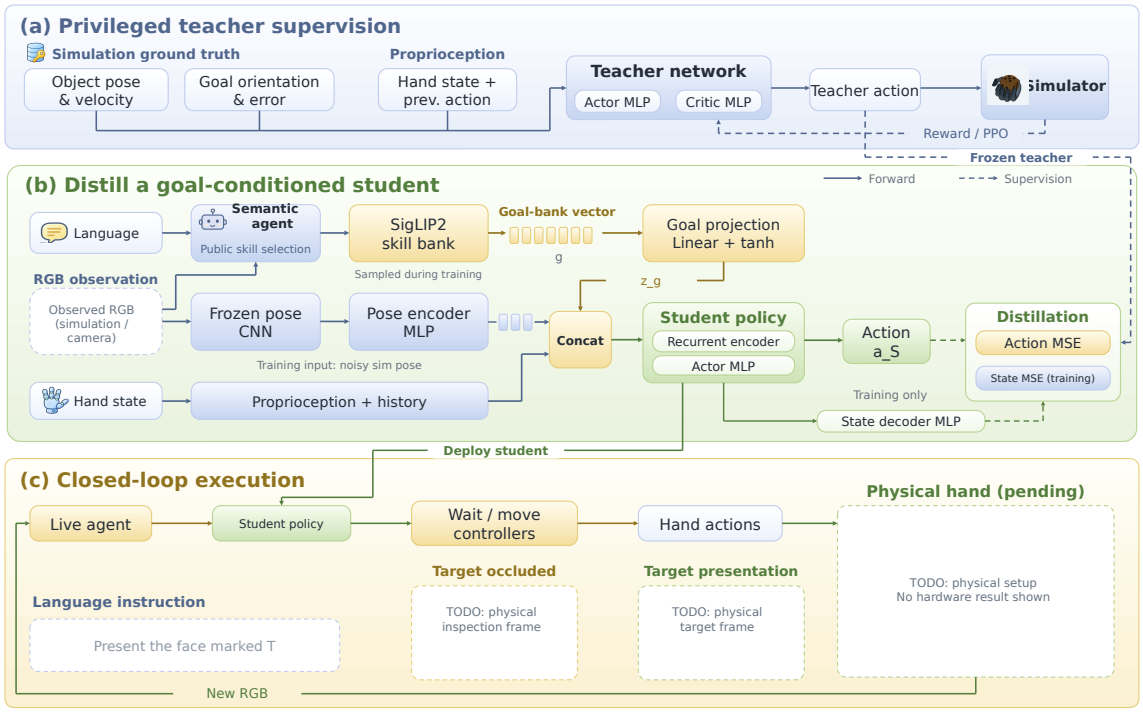


Figure 2. **REVEAL training and deployment pipeline.** Privileged supervision trains a student conditioned on a public goal-bank vector. The semantic agent selects skills, and local controllers adapt their execution. Dashed physical-hand slots are pending; they do not report hardware results.

4.2. Semantic grounding and public actions

The cube uses six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [19]; each is bound to a color skill and canonical asset orientation.

Semantic agent interface. Given the requested symbol and current image, the hosted agent emits a validated public action (inspect, reorient, verify, stop, or abstain). A color argument selects a vector from B; inspection may change the motor goal without binding the requested symbol. Agent-selected inspection/reorientation actions have bounded 10–90-step durations. Request latency is separate from action duration; updated images and public action history inform subsequent decisions. Public arguments map deterministically to the student goal; hidden agent features are not used.

4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector s_t from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal g_t is encoded separately. Before a

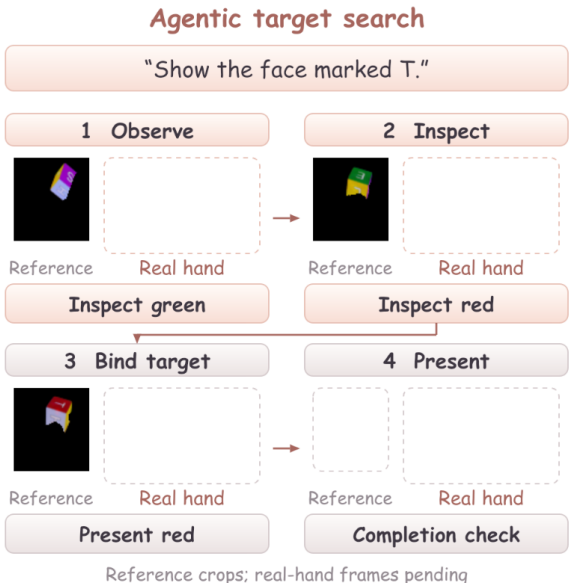


Figure 3. **Visual target search.** The agent observes, inspects candidate faces, binds the requested target, and issues a presentation command. The reference crops are simulated; real-hand frames are pending.

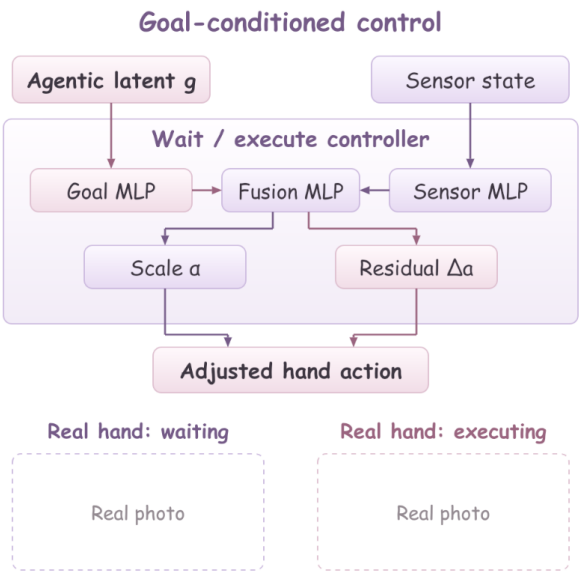


Figure 4. **Local control component.** A goal-conditioned actor combines sensor and goal features to scale the student proposal and add a bounded residual. The photo slots in the source illustration are pending and do not show real-hand results.

binding is established it denotes the startup or inspection skill, not the unavailable future target.

A sensor encoder and a goal encoder feed a fusion layer. The waiting actor predicts a speed logit r_t and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero outputs reproduce fixed slowing with coefficient 0.15 while a query is pending. The environment's joint mapping and smoothing remain unchanged. The residual permits contact adjustments outside the interpolation between current joints and the student proposal.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

The waiting-actor refit changes only pending-request outputs; the student, predictor, execution controller, and stopping gate remain fixed. Training details are provided in the supplement.

4.4. Learning execution after a decision

When $m_t = 0$, a separate execution actor supplies v_t in Eq. 6, using the student proposal as its base. It learns a bounded teacher-supervised correction for non-pending execution. Its active goal is explicit, but the current evaluation does not establish the benefit of this separate goal-input pathway.

4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

Student adaptation and controller learning are evaluated separately because changing the student also changes the belief and action proposals received by the controllers.

5. Experiments

We organize evaluation around four questions. Q1 tests complete systems, Q2 studies local control, and Q3 tests visual feedback. Q4 specifies the pending hardware evaluation. Q2 and Q3 use only the marked cube.

Q1. How do complete systems compare across objects, mappings and instruction variants?
Q2. How should the hand act during deliberation?
Q3. Does continued visual feedback help?
Q4. Does the system work on physical objects?

5.1. Setup and reporting

Experiments use Revo3 assets in Isaac Lab 3.0/Newton with validated contact capacities. Unless specified, each condition has two evaluation seeds and 40 scheduled trials. Physical tasks are paired within each study. Cube systems share the frozen visual pose estimator, goal bank, latent student, 0.16-radian tolerance and measured-pose stopping. Other objects use separate students and noisy simulator pose; REVEAL additionally uses calibrated completion correction. Bare VLM baselines omit the added control actors and use a fixed unreadable-view schedule, while REVEAL adapts its inspection. REVEAL denotes the complete deployed system. Natural model latency and these backend differences are part of the complete-system comparison.

The latest cube release replaces only its waiting actor. Q1 and Q3 rates use common service-eligible slots across com-

pared methods; Q2 reports all 120 outcomes per method. Q1 and Q2 use distinct evaluation cohorts: Q1 measures complete systems across mappings, whereas Q2 isolates pending-request control on the cube. Their absolute rates are not matched estimates; controller gains are computed within Q2. The three-seed Q2 table is descriptive. Paired tests and any multiplicity corrections are specified separately for each study in the supplement.

5.2. Q1: Systems across objects and mappings

We compare original and changed semantic mappings across six object-specific releases. Historical native-image cohorts remain separate in the supplement; their REVEAL–fine-tuned differences are unresolved (cube $p = 0.267$, die $p = 0.4545$).

Aggregate system performance. Over 1,440 scheduled trials, equal-object mean success is 70.4% on original mappings and 65.8% on changed mappings. The strongest baseline means are 44.2% (fine-tuning) and 25.4% (rules), yielding gains of 26.2 and 40.4 percentage points. These post-hoc descriptive means weight six service-eligible object rates equally; they neither pool trials nor add a significance test.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials across two mappings. REVEAL has the highest observed rate in each cell (Table 1); these are object-specific systems, not common-policy transfer. Three historical service failures are excluded symmetrically. Milk-carton REVEAL drops in 15/40 original and 20/40 changed-mapping trials, so higher completion does not imply stable grasping. Reset and sensitivity audits are in the supplement.

Visual-semantic intervention. Only the upper image changes; predictor input and lighting remain fixed. Identity, cyclic letter–color permutation and masking use 40 matched states per method, all with reconstructed face images. REVEAL retains 32/40 successes after permutation; fine-tuning falls from 30/40 to 16/40 (Holm $p = 0.0130$, six contrasts); rules remain at 23/40. Masking yields 0/39, 3/40 and 0/40, respectively (full supplement table). Nine of eleven fine-tuned wrong bindings after permutation select the old association, consistent with learned shortcuts but not an exclusive memorization mechanism. One permutation does not establish broad generalization or equivalence.

Dice layout transfer. A separate 240-trial study rotates only the native visual mesh, preserving collision, inertia and the prompt distinguishing printed counts from motor-skill IDs. Table 1 reports 40 matched trials per cell, with no exclusions. Permuted-layout REVEAL contrasts give exploratory paired $p = 2.98 \times 10^{-8}$ against fine-tuning and $p = 5.96 \times 10^{-8}$ against rules; its layout contrast ($p = 0.688$) does not establish equivalence. REVEAL also receives probes, history and noisy feedback, so this supports system-level transfer, not isolated model superiority.

Language and wording coverage. Three object-specific cohorts test English/Chinese standard and rewritten instructions: three evaluation seeds, 60 scheduled trials per cell, 2,160 total. Table 2 uses common service-eligible slots; weights, mappings and scoring remain fixed, without Chinese retraining. The rule baseline also localizes its task/output instructions, so this tests deployed-system coverage. Rates are conditional on service availability; the supplement retains raw outcomes.

REVEAL reaches 67.8–76.7% on duck and 73.3–76.7% on hedgehog, exceeding fine-tuning in all eight conditions (Holm-adjusted paired $p \leq 0.0258$ across 36 method contrasts). Against rules, six of eight contrasts survive correction; English-standard differences remain unresolved. On cube, REVEAL leads numerically in three conditions, while fine-tuning leads on Chinese rewriting (81.1% versus 67.9%); none of their four contrasts is significant (all adjusted $p = 1.0$). Strict pairing sensitivity preserves these decisions. No language/wording contrast survives its separate 36-test correction; this does not establish language equivalence.

5.3. Q2: Controller ablations on the cube

We compare four ways of acting while a semantic request is pending: continuing the student proposal, fixed slowing with coefficient 0.15, learned slowing, and learned slowing with a bounded joint correction. The ablation changes pending-request control while retaining the same execution controller outside pending intervals. Each method is evaluated for 120 trials, with 40 trials per seed across three seeds.

Slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning the speed alone yields 82/120 successes (68.3%), below fixed slowing. These results favor combining speed adaptation with joint correction in this evaluation; failure stages describe when a drop occurs rather than identifying its cause. Seed settings and outcome breakdowns are reported in the supplement.

5.4. Q3: Visual feedback on the cube

Three separate paired cohorts test continued feedback, fresh images, and inspection policy (Table 4). Outcomes are not pooled.

Object	Motor pose	$n_{\text{orig}}/n_{\text{changed}}$	Original mapping			Changed mapping		
			Rule	Fine-tuned	REVEAL	Rule	Fine-tuned	REVEAL
Cube	RGB pose	40 / 40	57.5	75.0	**80.0**	57.5	40.0	**80.0**
Die	Noisy pose	40 / 40	32.5	57.5	**70.0**	2.5	0.0	**65.0**
Duck	Noisy pose	38 / 40	39.5	36.8	**65.8**	22.5	5.0	**60.0**
Hedgehog	Noisy pose	39 / 40	43.6	33.3	**71.8**	25.0	5.0	**80.0**
Pig	Noisy pose	40 / 40	52.5	40.0	**92.5**	37.5	7.5	**82.5**
Milk carton	Noisy pose	40 / 40	32.5	22.5	**42.5**	7.5	2.5	**27.5**
Mean	Object-specific	–	43.0	44.2	**70.4**	25.4	10.0	**65.8**

Table 1. **Q1: six objects and two semantic mappings.** Completion (%) uses common service-eligible trials; each cell has 40 scheduled trials. Mean weights six objects equally. Bold denotes the highest observed rate, not significance. Object-specific weights remain fixed across mappings; bare baselines omit added control modules. Cube uses reconstructed upper images and RGB pose; die rotates its visual mesh; other objects permute surface letters with noisy simulator pose. Native-image and language cohorts remain separate.

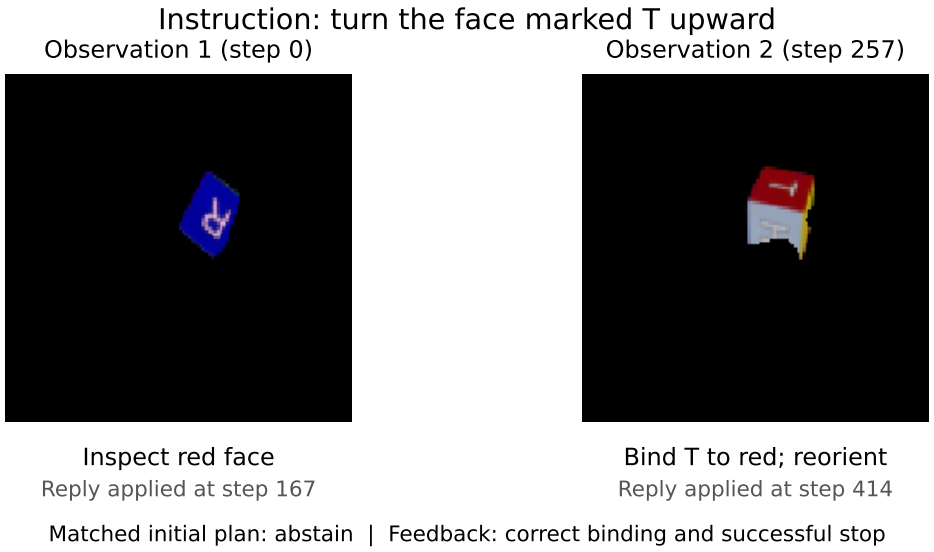


Figure 5. **Recorded inspection before binding.** Unretouched model-input frames from a recorded simulated trial. The initial plan abstains; feedback reveals T, then REVEAL binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from identical state and image. This selected example illustrates evidence acquisition, not wrong-binding correction or hardware performance.

(a) Is continued feedback useful? The initial-plan arm commits to up to six bounded actions from one RGB observation and measured feedback; full feedback allows up to six calls with updated images and history. Both retain the same local controllers, stopping gate and horizon. Success rises from 6/39 to 33/39 eligible trials: 27 pairs favor feedback and none favor initial planning (exact $p = 1.49 \times 10^{-8}$). One initial-plan service failure is excluded symmetrically. Of 28 eligible initial-plan abstentions, 22 succeed with feedback; 32 feedback trials first bind after inspection. This supports evidence acquisition, not correction of an executed wrong binding. The call budgets differ.

(b) Does the next image matter? Only upper-agent RGB changes: current images versus the exact first frame on every call. Both retain the six-call cap, trigger policy, live numeric feedback, history and local control. Success is 32/40 versus 9/40, with all 23 discordant pairs favoring fresh images (exact $p = 2.38 \times 10^{-7}$). Correct terminal bindings number 34 versus nine. Both arms first bind correctly on call one in the same nine trials; fresh images yield 25 additional correct bindings later, of which 23 succeed. Drops differ by only one. These post-hoc traces support acquiring target evidence; they do not measure first recognition accuracy or initial visibility.

(c) Does adaptive inspection help? Forty new pairs compare model-selected inspection directions and durations with a fixed color sweep of 90-step actions. Both arms retain the same model, controllers, fresh-image/history inter-

System	English Standard	English Rewrite	Chinese Standard	Chinese Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
REVEAL	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
REVEAL	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
REVEAL	**73.3**	**76.7**	**75.0**	**76.7**

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

Waiting policy	Success	%	Wait drops	Exec. drops	Other
Continue student	73/120	60.8	24	14	9
Fixed slowing	88/120	73.3	13	8	11
Learned slowing	82/120	68.3	7	14	17
Slowing + correction	**104/120**	**86.7**	**5**	**4**	**7**

Table 3. **Q2: controller ablation on the cube.** Each method has 120 trials across three seeds. Wait and execution drops denote failure stages; Other contains the remaining failures. Outcome counts sum to 120 in each row.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98
(c) Inspection policy: same model and controllers				
Fixed sweep	31/40	77.5	1	126
Adaptive inspection	38/40	95.0	0	117

Table 4. **Q3: feedback, evidence and inspection on the cube.** Each panel schedules 40 trials per arm. (a) excludes one service-affected pair and reuses Q2's initial full-feedback subset. (b) changes only upper RGB. (c) changes inspection direction/duration. (b,c) have no exclusions. Drops/calls are raw totals; cohorts and realized compute differ.

face and stopping rule; actual executed actions enter feedback. The model still decides whether to inspect, bind or abstain. Adaptive inspection achieves 95.0% success versus 77.5%, a 17.5-point observed gain. Nine pairs favor adaptive and two favor fixed inspection (exact $p = 0.0654$), leaving the difference unresolved at $\alpha = 0.05$. Correct terminal bindings are 39 versus 32, with zero versus one drop. This is a favorable trend against one fixed sweep, not a general or statistically significant advantage.

5.5. Q4: Multi-object hardware evaluation

Hardware protocol. **[TODO: Complete the multi-object physical-hand evaluation.]** No hardware outcomes are available yet. The planned comparison uses REVEAL and two baselines per object, reporting completion, drops, false stops and inference cost. Calibration, resets, independent scoring and synchronized-video requirements are specified in the supplement.

Supporting cube perception study. On 6,144 rendered observations, RGB-D reaches 64.73% pose recall versus 37.53% for RGB at 10 mm/10°. Ideal masks and virtual depth limit this estimator comparison; downstream gains and physical-camera accuracy remain untested. The supplement details its protocol and occlusion analysis.

6. Discussion and Limitations

Evidence boundaries. Fresh images aid cube completion; the adaptive-inspection advantage remains statistically unresolved. Q1 confounds semantic models with inspection, control and stopping. Two languages and templates do not establish broad multilingual generalization or equivalence.

Unresolved controls. Two aggregate-only Q2 seeds prevent seed-level variability estimates and paired tests. Explicit controller goals have no demonstrated benefit; comparison with a generic time-aware residual controller remains open. Q2/Q3 mechanisms are cube-only, and matched call caps do not equalize realized compute.

Deployment scope. Object-specific weights, rendered cube RGB and noisy simulator pose on other objects do not establish real-camera robustness or zero-shot transfer. Only one changed mapping is tested per object. Full-quaternion scoring fixes a twist unspecified by the instruction. Hosted-model revisions limit replication; service-conditional rates omit end-to-end availability. Hardware evaluation remains pending.

7. Conclusion

REVEAL connects visual search and semantic binding to continuous in-hand control. Six-object mean success is 70.4% on original mappings and 65.8% on changed mappings. Cube studies support fresh observations and waiting-time joint correction; physical-hand deployment remains unvalidated.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

References

[1] Marcin Andrychowicz, Bowen Baker, Maciek Chociej, Rafal Józefowicz, Bob McGrew, Jakub Pachocki, Arthur Petron, Matthias Plappert, Glenn Powell, Alex Ray, Jonas Schneider, Szymon Sidor, Josh Tobin, Peter Welinder, Lilian Weng, and Wojciech Zaremba. Learning dexterous in-hand manipulation. *The International Journal of Robotics Research*, 39(1): 3–20, 2020. 2

[2] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, Mei Li, Kaixin Li, Zicheng Lin, Junyang Lin, Xuejing Liu, Jiawei Liu, Chenglong Liu, Yang Liu, Dayiheng Liu, Shixuan Liu, Dunjie Lu, Ruilin Luo, Chenxu Lv, Rui Men, Lingchen Meng, Xuancheng Ren, Xingzhang Ren, Sibo Song, Yuchong Sun, Jun Tang, Jianhong Tu, Jianqiang Wan, Peng Wang, Pengfei Wang, Qiuyue Wang, Yuxuan Wang, Tianbao Xie, Yiheng Xu, Haiyang Xu, Jin Xu, Zhibo Yang, Mingkun Yang, Jianxin Yang, An Yang, Bowen Yu, Fei Zhang, Hang Zhang, Xi Zhang, Bo Zheng, Humen Zhong, Jingren Zhou, Fan Zhou, Jing Zhou, Yuanzhi Zhu, and Ke Zhu. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025. 2

[3] Arjun Bhardwaj, Mayank Mittal, Maximum Wilder-Smith, Vaishakh Patil, and Marco Hutter. GoalDex: Learning image-goal-conditioned policies for dexterous in-hand manipulation. *arXiv preprint arXiv:2611.02171*, 2026. 2

[4] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026. 2

[5] Kevin Black, Manuel Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, pages 33383–33407, 2025. 2

[6] Ning Chen, Yankai Fu, Junkai Zhao, Qianpu Sun, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. LiMA: Bridging long-term imagination to real-time dexterous manipulation via asynchronous diffusion. *arXiv preprint arXiv:2609.28431*, 2026. 2

[7] Tao Chen, Jie Xu, and Pulkit Agrawal. A system for general in-hand object re-orientation. In *Proceedings of the 5th Conference on Robot Learning*, pages 297–307, 2022. 2

[8] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D value maps for robotic manipulation with language models. In *Proceedings of the 7th Conference on Robot Learning*, pages 540–562, 2023. 2

[9] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *Proceedings of the 6th Conference on Robot Learning*, pages 1769–1782, 2023. 2

[10] Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. ReKep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *Proceedings of the 8th Conference on Robot Learning*, pages 4573–4602, 2025. 2

[11] Brian Ichter, Anthony Brohan, Yevgen Chebotar, Chelsea Finn, Karol Hausman, Alexander Herzog, Daniel Ho, Julian Ibarz, Alex Irpan, Eric Jang, Ryan Julian, Dmitry Kalashnikov, Sergey Levine, Yao Lu, Carolina Parada, Kanishka Rao, Pierre Sermanet, Alexander T. Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Mengyuan Yan, Noah Brown, Michael Ahn, Omar Cortes, Nicolas Sievers, Clayton Tan, Sichun Xu, Diego Reyes, Jarek Rettinghouse, Jornell Quiambao, Peter Pastor, Linda Luu, Kuang-Huei Lee, Yuheng Kuang, Sally Jesmonth, Nikhil J. Joshi, Kyle Jeffrey, Rosario Jauregui Ruano, Jasmine Hsu, Keerthana Gopalakrishnan, Byron David, Andy Zeng, and Chuyuan Kelly Fu. Do as i can, not as i say: Grounding language in robotic affordances. In *Proceedings of the 6th Conference on Robot Learning*, pages 287–318, 2023. 2

[12] Kailin Li, Puhao Li, Tengyu Liu, Yuyang Li, and Siyuan Huang. ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6991–7003, 2025. 2

[13] Sizhe Lester Li, Evan Kim, Xingjian Bai, Tong Zhao, Tao Pang, Max Simchowitz, and Vincent Sitzmann. Turning video models into generalist robot policies. *arXiv preprint arXiv:2605.27817*, 2026. 2

[14] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 9493–9500, 2023. 2

[15] Haozhi Qi, Ashish Kumar, Roberto Calandra, Yi Ma, and Jitendra Malik. In-hand object rotation via rapid motor adaptation. In *Proceedings of the 6th Conference on Robot Learning*, pages 1722–1732, 2023. 2

[16] Haozhi Qi, Brent Yi, Sudharshan Suresh, Mike Lambeta, Yi Ma, Roberto Calandra, and Jitendra Malik. General in-hand object rotation with vision and touch. In *Proceedings of the 7th Conference on Robot Learning*, pages 2549–2564, 2023. 2

[17] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth Inter-*

national Conference on Artificial Intelligence and Statistics, pages 627–635, 2011. 3

[18] Kohei Sendai, Maxime Alvarez, Tatsuya Matsushima, Yutaka Matsuo, and Yusuke Iwasawa. Leave no observation behind: Real-time correction for VLA action chunks. *arXiv preprint arXiv:2509.23224*, 2025. 2

[19] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 4

[20] Jianwei Yang, Reuben Tan, Qianhui Wu, Ruijie Zheng, Baolin Peng, Yongyuan Liang, Yu Gu, Mu Cai, Seonghyeon Ye, Joel Jang, Yuquan Deng, and Jianfeng Gao. Magma: A foundation model for multimodal AI agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14203–14214, 2025. 2

[21] Max Yang, Chenghua Lu, Alex Church, Yijiong Lin, Christopher J. Ford, Haoran Li, Efi Psomopoulou, David A. W. Barton, and Nathan F. Lepora. AnyRotate: Gravity-invariant in-hand object rotation with sim-to-real touch. In *Proceedings of the 8th Conference on Robot Learning*, pages 4727–4747, 2025. 2

[22] Brian Zhu, Momen Khalil, E Harrison, Emanuele Poggi, Philipp Schmitt, Bernd Kast, Philine Meister, Pranav Atreya, Qiyang Li, Finn Ferchau, Cesar Colmenero, Yash Shahapurkar, Gokul Narayanan, Melih Erdogan, Kai Wurm, Georg von Wichert, Oier Mees, Eugen Solowjow, Andrew Wagenmaker, and Sergey Levine. Learning to act while waiting: RL finetuning of generalist robot policies under inference latency. *arXiv preprint arXiv:2608.23831*, 2026. 2

[23] Yixiang Zhu, Yonghao Chen, Jiaxiang Zou, Yusong Hu, and Xinyu Chen. DEFLECT: Temporal counterfactual preference learning for delay-robust asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. 2